

today's obj/dif:

- Formulate relation H-T lemma 2.3 to φ sat.

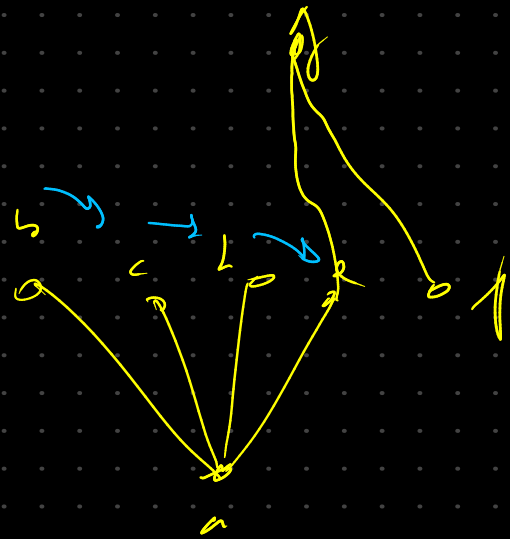
let's assume we have a weak-Pancari-tutte

$$\begin{aligned}\varphi(bc) &= 1 \\ \varphi(bf) &= 0 \\ \varphi(cf) &= 1\end{aligned}$$

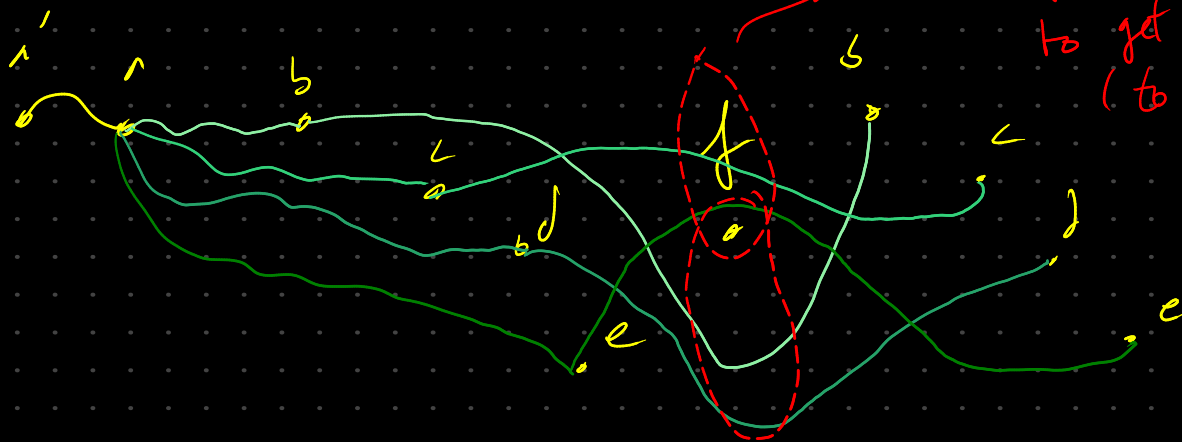
also

$$\varphi(df) = 0$$

$$\varphi(ef) = 1$$

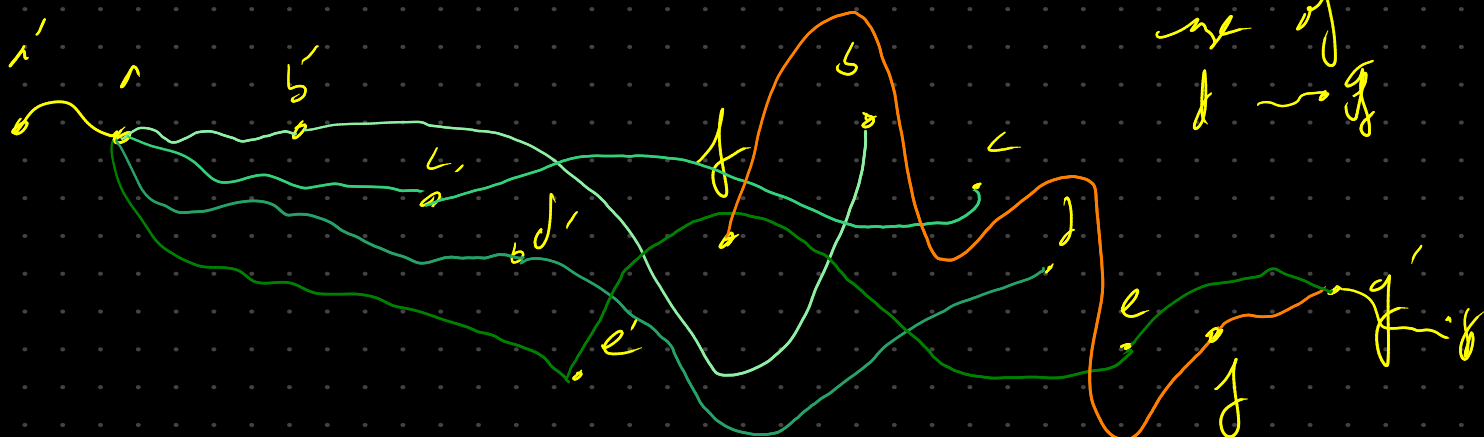


not letting the green edges to get "straighter" (to be cross-free)

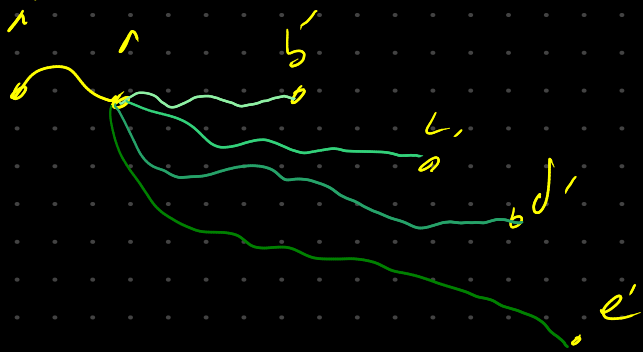


take also

we of $d \rightarrow g$

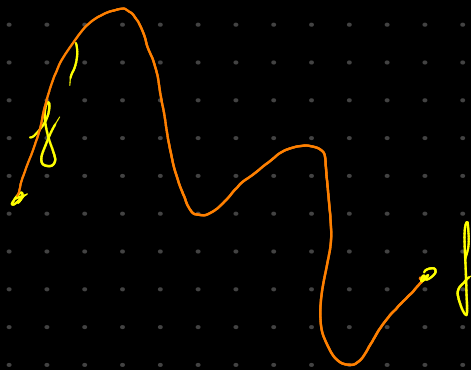
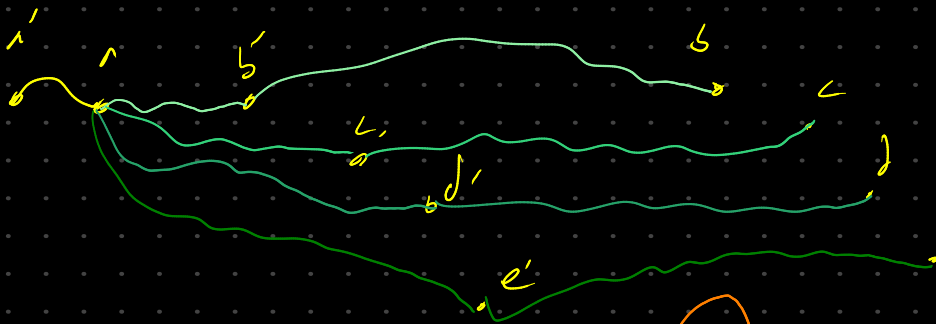


if we start the construction from the left



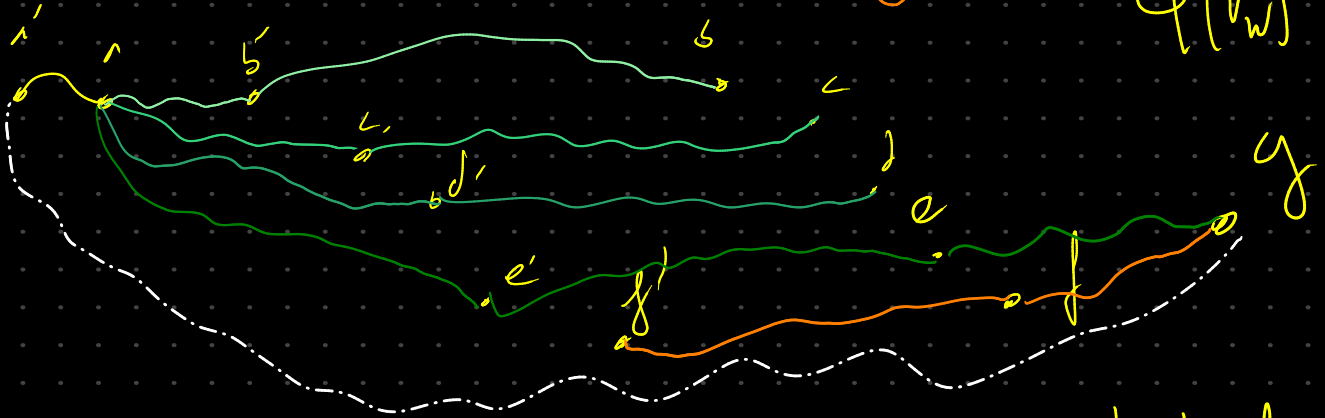
$$f(v) = x, \quad x \in \{0, 1\}$$

$f \in C_{\text{green}}$

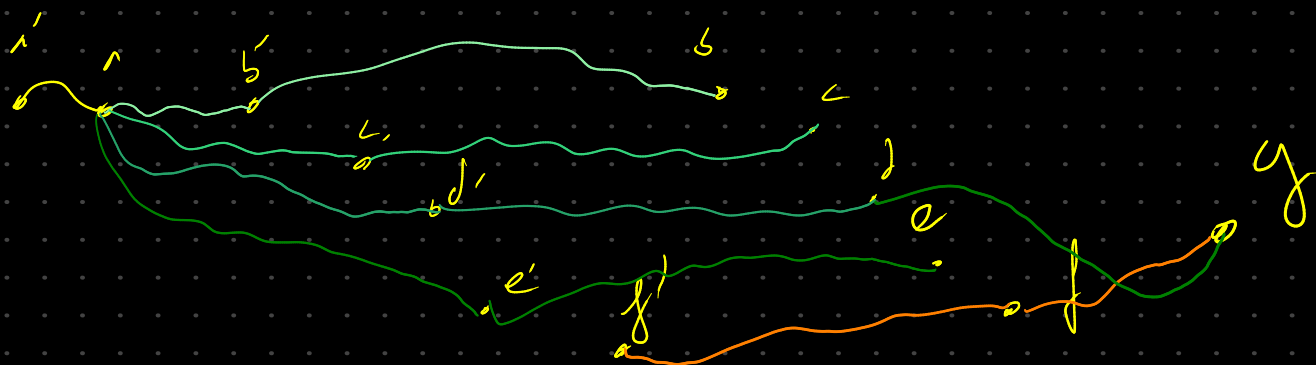


$$f(v) = f(w)$$

$f \in C_{\text{green}}$
 $f \in C_{\text{orange}}$

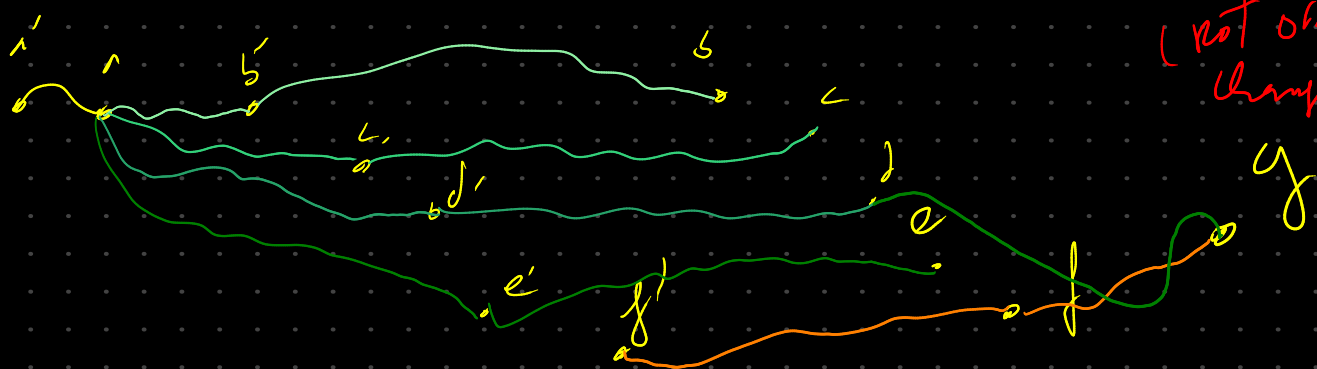


what if we had $(b, g) \in E(w)$ instead of (e, g)

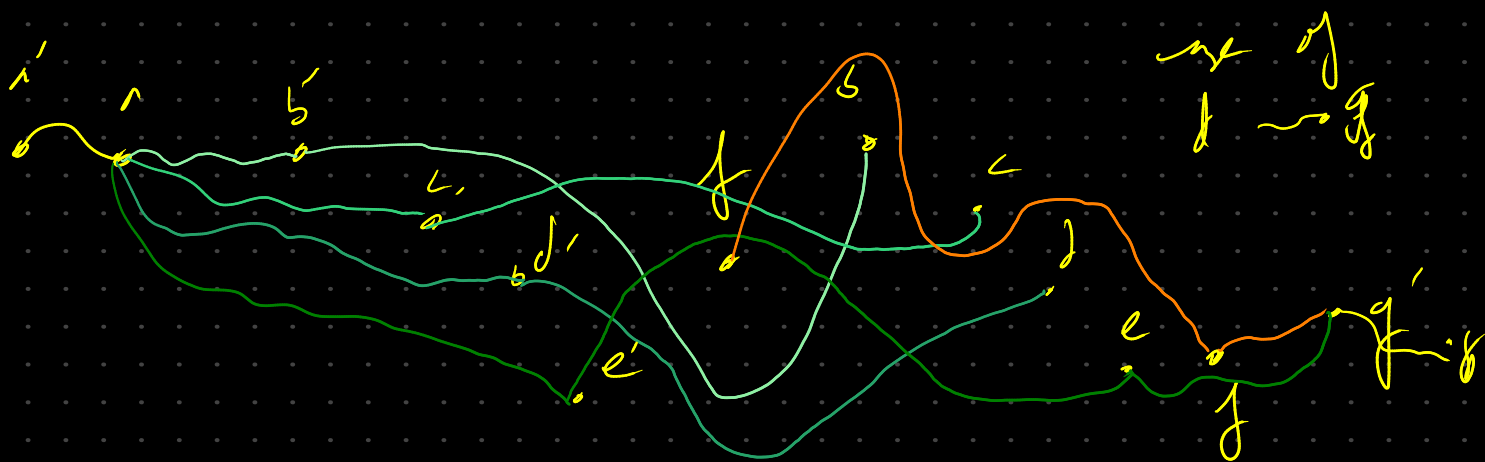


we remove the old mapping $\ell'(bf) = f(g)$

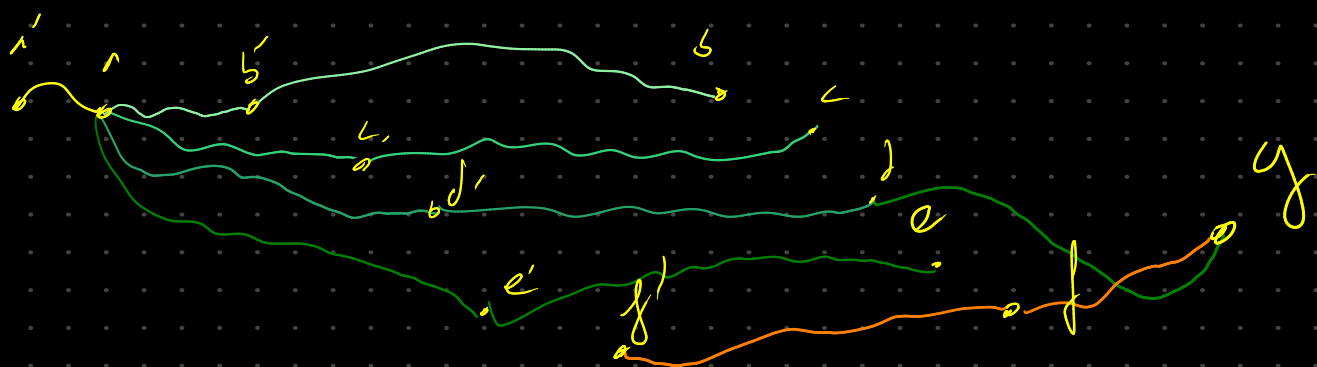
(not order
change)



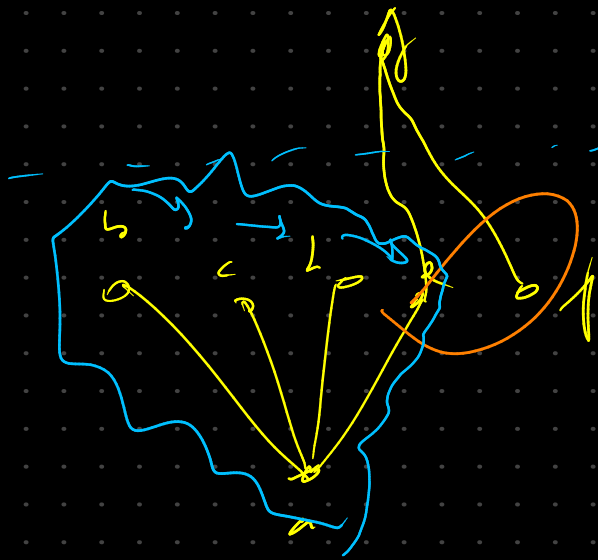
what if $\ell(lf) = \ell(ej) = 0$



we do
l → g



the order depends on what's the order of both camps in the set of where they intersect.



defines the order of H_g in regard to H_n

H_n

for each level we keep the trade of the connected camps

if in a level we connected to camps, order of the vertices from the diff camp will be the same as the one with the rotation in the common endpoint

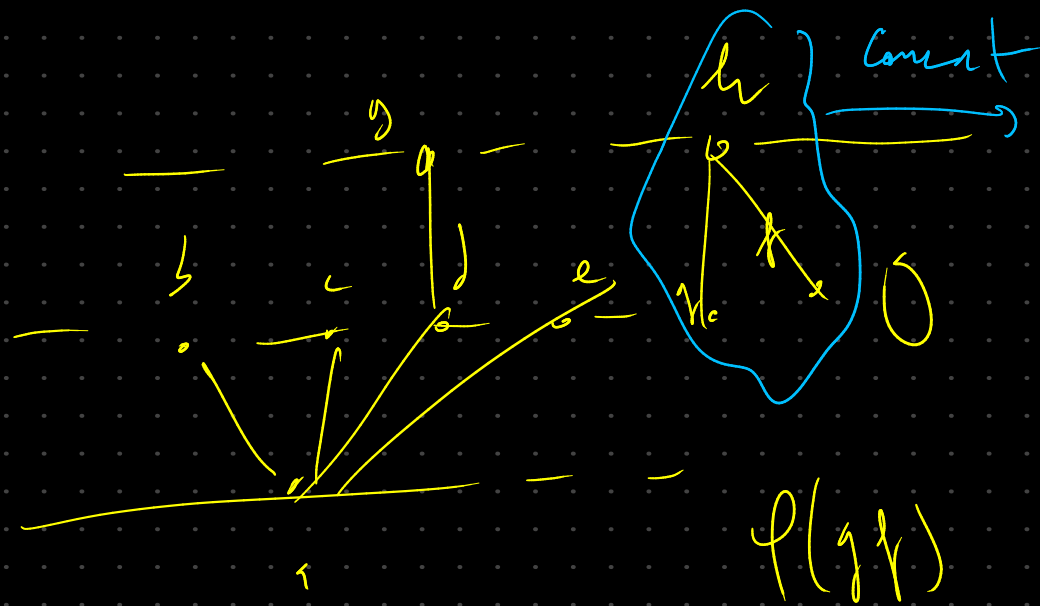
$$H_1 = \{a\} \quad H_2 = \{b\} \quad H_3 = \{c\}$$



$$H_1 = \{a, b, c, d, e\}$$

$$\varphi(b, c), \varphi(c, d), \varphi(d, e)$$

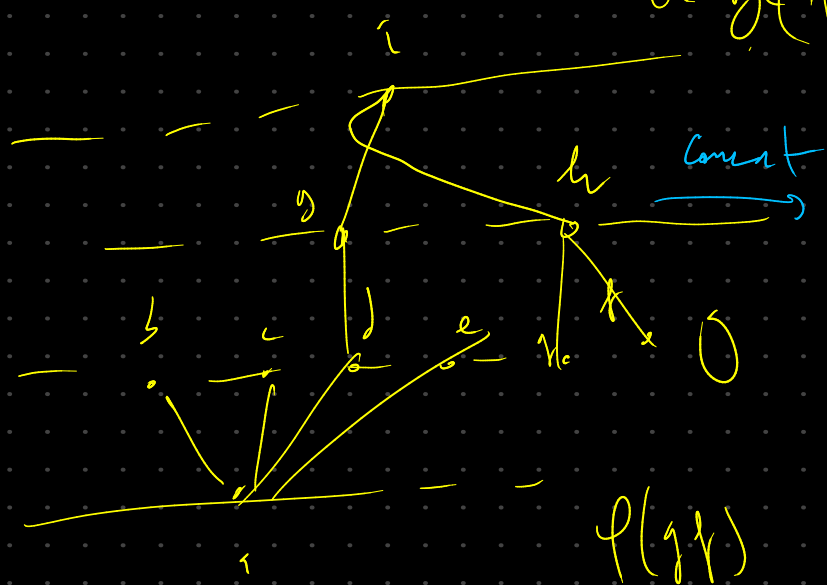
$$\varphi(b, c), \varphi(c, d), \varphi(d, e)$$



All $H_2, H_3 \rightarrow H_3$
 will have
 the same
 relative order
 btw each other

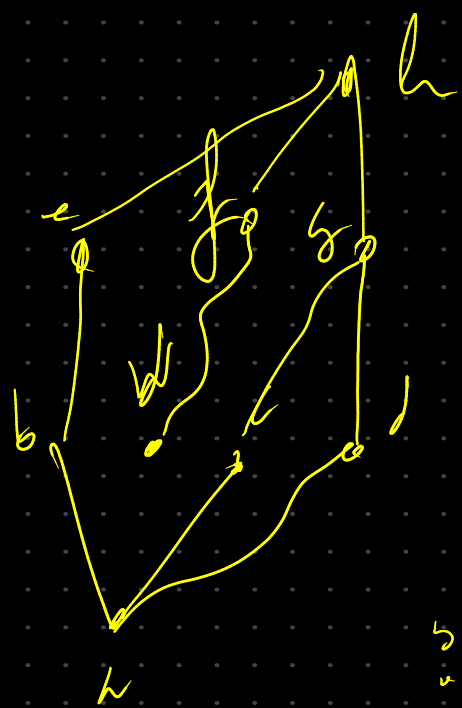
$$\phi(g_1)$$

$$\text{Merge}(H_1, H_{23})$$



$$\phi(H_1, H_{23}) = 0$$

$$\phi(g_1)$$



$h+1$ adjacent arcs (insert no cycles)
 no crossing

$$\phi(b_1) = \phi(c_1) = 2\phi(a_1)$$

change are (the one that are flexible)

$$H_1 = \{a\}$$

heap
 state
 of free elements
 (how?)



Change of flow: greedy pick assign. to eq. class



$$\varphi(b_c) = \varphi(c_s)$$

$$= \wedge$$

$$= h$$

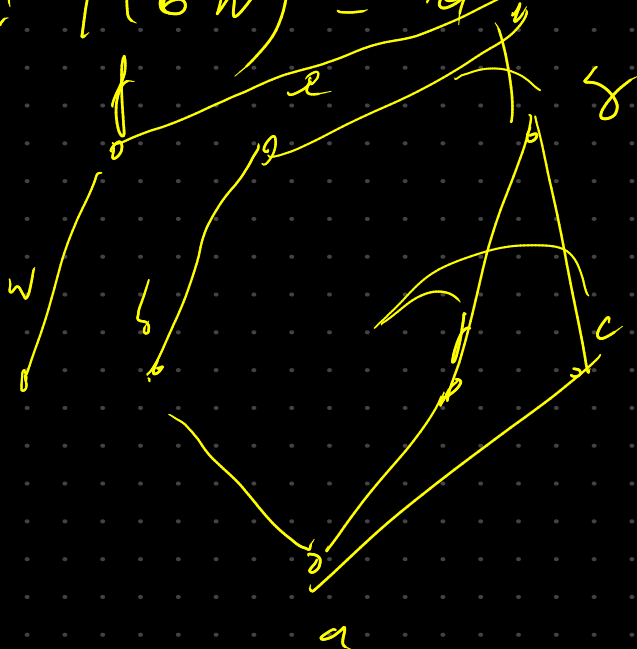
$$\begin{aligned} \varphi(f_o) \\ = \varphi(w_c) \\ \varphi(w_d) \end{aligned}$$

look for adjant edges with endpoints
in eq. class vertices

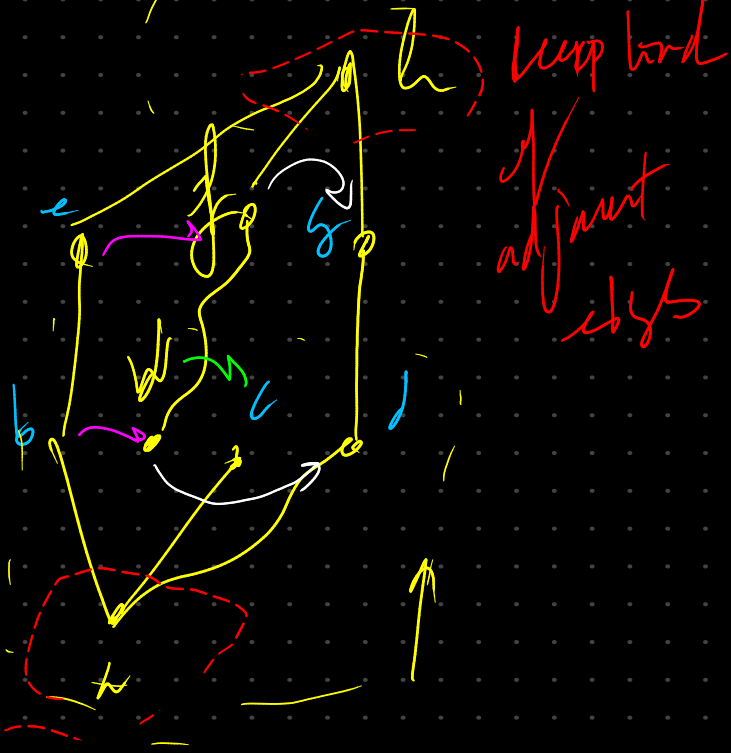
$$\varphi(b_g) = \varphi(w_d) = \varphi(w_f) = \varphi, \varnothing$$

red: $\varphi(c_d) = 0$

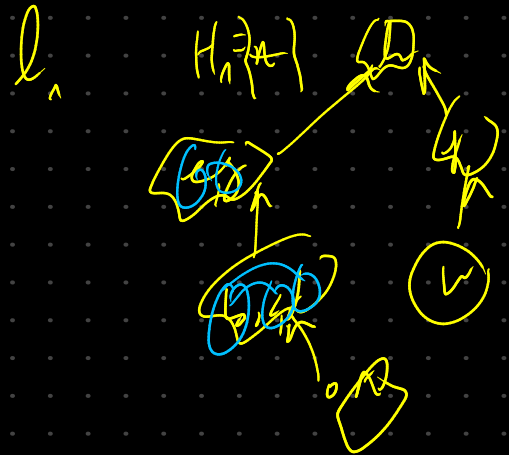
purple: $\varphi(b_w) = \varnothing, h$



but we have an issue with this logR.

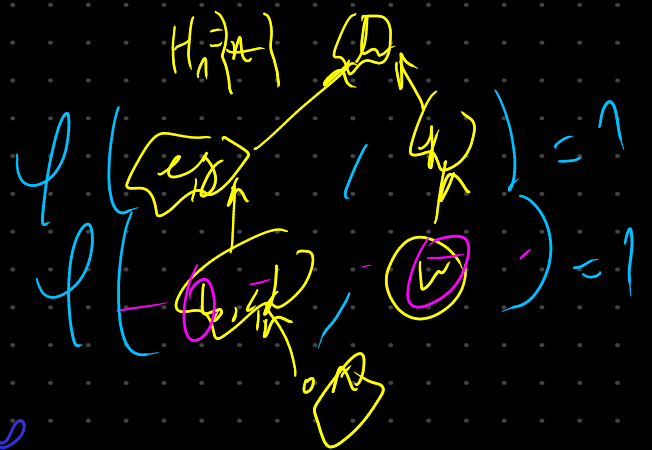


$l_1 \quad H_1 = \{a, b, c, d, e, f, g, h\}$
 $l_2 \quad H_1 = \{a, b, c, d, e, f\} \quad H_2 = \{g, h\}$
 $l_2 \quad H_1 = \{a, b, c, d\} \quad H_2 = \{e, f\}$



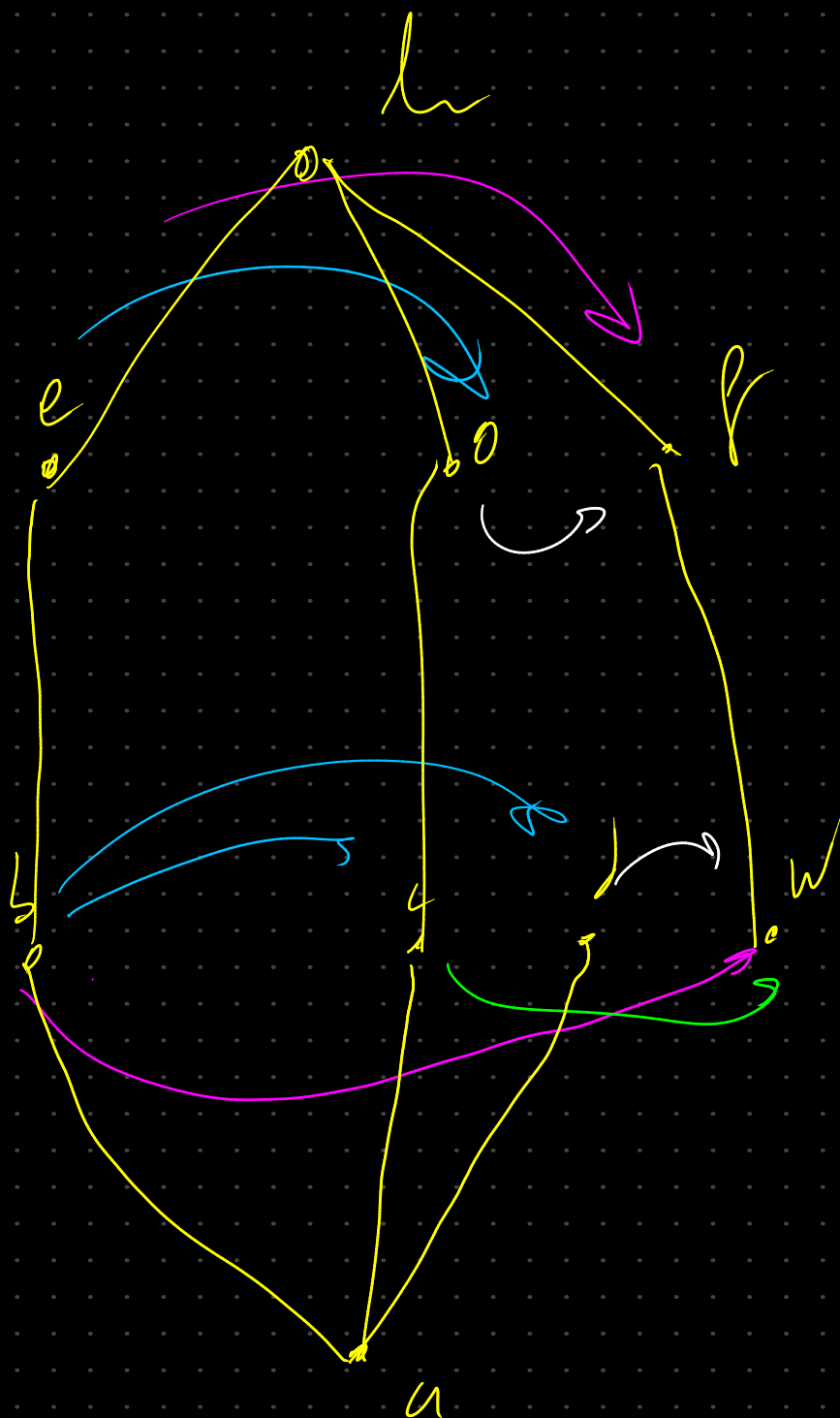
$$\varphi(b, c) = \varphi(b, d) = \varphi(e, g) = 1$$

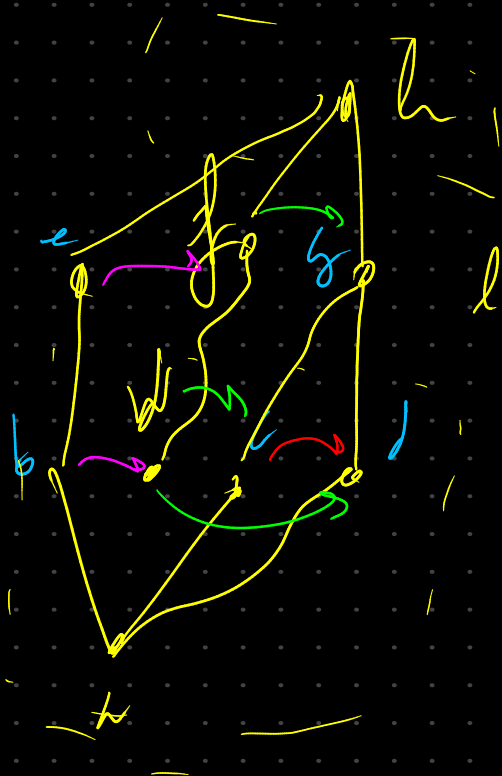
$$\varphi(b, w) = \begin{cases} 0 \\ 1 \end{cases} = b$$



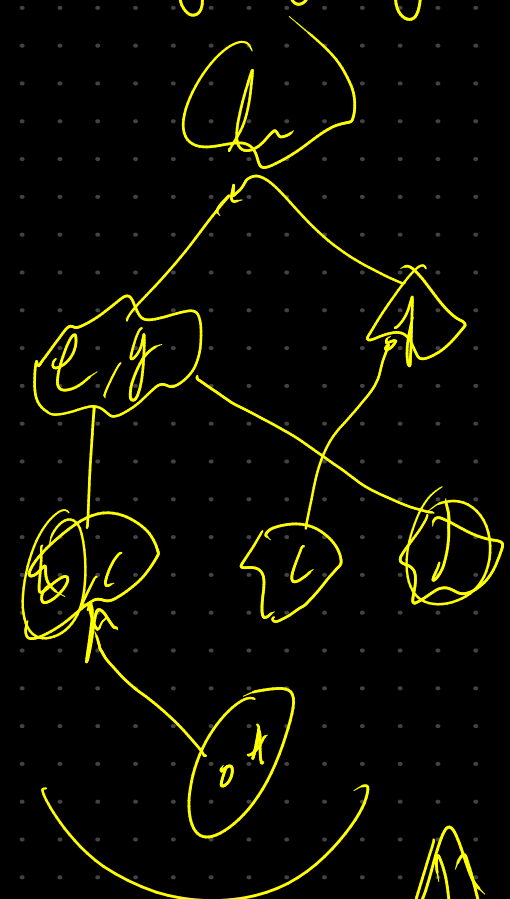
$$\left. \begin{aligned} \varphi(b, w) &= b \\ \varphi(c, w) &= b \\ \varphi(d, w) &= b \end{aligned} \right\} \varphi(e, f) = b$$

$$L_1 = 1 \quad L_2 = 1 \quad L_3 = 1 \quad L_n = 1$$





h
wouldn't do anything



easy to solve
(no adjacent edges)

$$p(b, 18) \rightarrow p(h, h) = \wedge$$